

Nathan Jacobson

COMPUTER SCIENCE STUDENT · SOFTWARE DEVELOPER

📍 Vancouver, BC, Canada

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🏠 <https://nathanjacobson.ca>

🌐 grqphical

📄 Nathan Jacobson

Skills

Programming Languages Go, Python, C, C++, HTML/CSS/JS, SQL, Lua
Libraries/Frameworks FastAPI, Numpy, Pandas, OpenGL, Vue.js, ROS2
Technical Skills Linux, Docker, Vim, CMake

Technical Experience

Software Developer

Burnaby, BC

SFU RACERBOT DESIGN TEAM

January 2026 - Present

- Developed the software for a 1/10 scale autonomous race car, based on the University of Pennsylvania's RoboRacer platform
- Built a collision-detection algorithm using C++ and ROS 2 that uses data from the car's LIDAR sensors, ensuring the car would not collide with any walls or obstacles
- Designed the team's website with HTML/CSS/JS, allowing our team to reach out to a wider audience

Projects

C to WebAssembly Compiler

🌐 grqphical/webc

DESIGNER/DEVELOPER

January 2026 - Present

- Built a compiler in Go that can compile C code into WebAssembly, allowing it to be run in the browser
- Designed a Pratt parser to generate an Abstract Syntax Tree, ensuring code compilation is accurate and swift
- Wrote a JavaScript port of the C standard library that is able to be both run in the browser and on the server with tools such as Node.js, ensuring maximum compatibility with existing C projects

Reverse Proxy

🌐 grqphical/interchange

DESIGNER/DEVELOPER

June 2025 - November 2025

- Created a reverse proxy server with Golang, allowing users to easily route HTTP traffic to various services
- Developed it to include the ability to proxy HTTP requests, host static files, and whitelist/blacklist IP addresses; giving users versatility in deploying their web applications

DND Character Creator

🌐 grqphical/CharacterForge

DESIGNER/DEVELOPER

February 2026 - Present

- Built a web app with Vue.js that allows users to create character sheets for DND 5th edition, giving DND players an easy way to create and manage their characters
- Designed the app to store everything using the browser's local storage ensuring that users can access their characters even without an internet connection

Work Experience

Tae Kwon Do Instructor

Port Moody, BC

TRANSCENDING MARTIAL ARTS

September 2025 - December 2025

- Instructed children aged 4-14 martial arts techniques such as strikes, blocks, patterns, and weapons training, resulting in improved student discipline and form
- Led a group of volunteer assistant instructors to aid with class management, creating a more positive and disciplined learning environment
- Developed structured lesson plans based on the class' age and skill levels, ensuring all students would be able to fully participate in the class

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Work Experience (cont.)

Information Officer

Port Moody, BC

ELECTIONS CANADA

April 2025

- Confirmed voter identification when they entered the polling station, ensuring voters had brought the correct forms of ID required to vote
- Directed voters to the correct stations and assisted voters with physical disabilities, allowing everyone to exercise their democratic rights without any barriers or obstacles
- Recorded vote counts after polls had closed in an efficient manner, allowing election results to be quickly compiled and released to the public

Relevant Coursework

- **Systems Programming (Jan. 2026 - Present):** Created a group chat server using C and TCP/IP and a multi threaded implementation of Map-Reduce.
- **Data Structures and Programming (Jan. 2026 - Present):** Built a Tic-Tac-Toe game as well as an artificial intelligence algorithm to play it

Education

Simon Fraser University

Burnaby, BC

BSc IN COMPUTER SCIENCE

September 2025 - Present

Current GPA: 3.6

Volunteer Experience

Assistant Tae Kwon Do Instructor

Port Moody, BC

TRANSCENDING MARTIAL ARTS

June 2022 - August 2023

- Aided instructors in teaching children aged 4-9 martial arts techniques
- Managed the class, ensuring students were focused
- Helped students who were struggling with following the lesson, ensuring they could fully understand the techniques being taught

Summer Camp Volunteer

Burnaby, BC

SIMON FRASER UNIVERSITY SUMMER CAMPS

July 2023

- Assisted instructors in running chemistry and engineering camps for children ages 5-6
- Supervised groups of children during structured activities, lunch breaks, and free time

Achievements

BC Achievement Scholarship

BRITISH COLUMBIA MINISTRY OF EDUCATION AND CHILD CARE

October 2025

Scholarship awarded to students who've demonstrated academic excellence through grades 10-12

Extra-Curricular Activities

Tae Kwon Do

TRANSCENDING MARTIAL ARTS

September 2017 - Present

- Currently hold a 1st degree black belt in Tae Kwon Do

Interests

- Photographing natural and urban landscapes
- Acting as a Dungeon Master for a weekly Dungeons & Dragons campaign
- Building and managing an NHL fantasy team